Systems Biology Integration of Glycomics and Cytoskeletal Transport Networks Reveals Microgravity-Induced Cell Wall Remodeling Mechanisms in Arabidopsis thaliana

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Abstract

Background: Spaceflight exposes biological systems to microgravity, inducing pronounced biomechanical stress and morphological reprogramming in plant cell walls. In NASA study OSD-615 (Advanced Plant EXperiment 03-1 / APEX-03-1), high-throughput glycome profiling across 155 monoclonal antibodies revealed extensive structural remodeling of non-cellulosic polysaccharides in *Arabidopsis thaliana* seedling roots grown aboard the International Space Station (ISS) Veggie facility. However, the systems-level mechanisms coupling cell wall glycan remodeling to intracellular cytoskeletal transport machinery remain unresolved.

Methods: Here, we present an integrative systems biology framework that links continuous glycomic epitope matrices from OSD-615 to companion spaceflight transcriptomics (APEX-03-2 / OSD-218 and OSD-217) and interactome topologies. We employed sparse Partial Least Squares (sPLS) regression, Weighted Gene Co-expression Network Analysis (WGCNA), and protein–protein interaction (PPI) network modeling (STRING/AraNet) to correlate 155 glycan epitope vectors with motor protein complexes (kinesins, myosins), microtubule-associated proteins (MAP65, SPR1, CLASP), actin-regulatory machinery (ARP2/3, formins, profilins), and cellulose synthase complexes (CSCs). Furthermore, we developed a stochastic Monte Carlo vesicle transport model simulating secretory flux from Golgi stacks to the cell cortex under 1g and microgravity conditions, and reviewed mass spectrometry workflows (EThcD, metabolic click-chemistry, lectin affinity) for mapping intracellular O-GlcNAcylation on motor complexes.

Results: Microgravity triggered significant alterations in cell wall glycomes, characterized by enhanced extractability of β-(1,4)-xylan epitopes (detected by CCRC-M138, CCRC-M139, CCRC-M140; +1.8 to +2.4 $\log_2$ FC) and dynamic redistribution of arabinogalactan proteins (JIM13, JIM14) and xyloglucans (CCRC-M1, CCRC-M88). Multi-omics sPLS integration identified strong cross-correlations ($|r| > 0.75$) between xylan epitope abundance and transcriptional activation of secondary wall cellulose synthases (*CESA4*, *CESA7*), glycosyltransferases (*IRX9*, *IRX10*), and motor components (*KIN12A*, *MYA1*). Stochastic simulations demonstrated that microgravity-induced cortical microtubule disorientation and actin cable fragmentation decrease vesicle arrival efficiency at the cell plate by 28%, directly altering matrix polysaccharide deposition rates.

Conclusion: These findings establish that microgravity-induced cell wall remodeling is mechanistically coupled to cytoskeletal motor dynamics and vesicular trafficking. We provide a FAIR-compliant open-access research repository, including an interactive 10-tab web dashboard, reproducible analysis pipelines, and full data dictionaries to support space agricultural engineering for extended lunar and Martian missions.

1 Introduction

Plants evolved under an invariant 1g gravitational vector on Earth, utilizing gravity as a primary cue for directional growth, mechanical reinforcement, and cellular morphogenesis[1,2]. In the microgravity environment of low Earth orbit (LEO), such as aboard the International Space Station (ISS), the absence of sedimentation forces fundamentally disrupts cellular mechanics, triggering extensive transcriptional reprogramming, cytoskeletal reorganization, and cell wall remodeling[3,4,5].

The plant cell wall is a dynamic, complex extracellular matrix composed of cellulose microfibrils embedded in an amorphous ground substance of non-cellulosic matrix polysaccharides (xyloglucans, xylans, mannans, pectins) and structural glycoproteins, including arabinogalactan proteins (AGPs) and extensins[6,7]. In growing root tissues, cell wall loosening, matrix synthesis, and directional expansion require continuous physical coordination between the intracellular cytoskeleton and extracellular polysaccharide deposition[8,9].

1.1 The Cytoskeleton–Secretory Vesicle Transport Nexus

In plant cells, unlike animal cells where long-range organelle transport relies predominantly on microtubule-dependent kinesin and dynein motors, long-range cytoplasmic streaming and post-Golgi vesicle trafficking are driven primarily by class XI myosin motors moving along filamentous actin (F-actin) cables [10,11]. Concurrently, cortical microtubules (CMTs) directly govern the trajectory and insertion of cellulose synthase complexes (CSCs) into the plasma membrane via Cellulose Synthase Interactive 1 (CSI1/POM2) linkers [12,13]. Furthermore, kinesin motor proteins (such as Kinesin-4/FRA1 and Kinesin-12/PAKRP1) operate along microtubule arrays during cell division, phragmoplast expansion, and secondary cell wall patterning, guiding secretory vesicles containing matrix polysaccharides and wall-modifying enzymes to target membrane domains [14,15].

When plants experience microgravity, cortical microtubule arrays frequently lose transverse alignment and become disoriented, while actin filaments exhibit altered bundling and aggregation [16,17]. This cytoskeletal disruption impairs the directional delivery of Golgi-derived matrix vesicles, resulting in aberrant cell wall assembly, altered wall elasticity, and root growth skewing [18,19].

1.2 Coupling Across Biochemical Axes: Intracellular Glycosylation vs. Cell-Surface Glycoproteomics

The functional connection between glycans and cytoskeletal machinery operates across two primary biochemical axes:

1. **Direct Intracellular Glycosylation (O-GlcNAcylation):** Unlike classic cell-surface complex *N*- and *O*-glycans, dynamic intracellular *O*-linked β-*N*-acetylglucosamine (*O*-GlcNAc) directly modifies nuclear and cytosolic proteins, competing with Ser/Thr phosphorylation [20,21]. Actin monomers, myosins, and regulatory proteins (such as α-actinin, fascin, and cofilin) undergo dynamic *O*-GlcNAcylation, regulating actin polymerization, stress fiber assembly, and motor kinetics [22,23]. Similarly, tubulin, dynein intermediate/light chains, dynactin p150$^{\mathrm{Glued}}$, and kinesin stalk domains are targets of *O*-GlcNAc transferase (OGT) in animal systems, and homologous enzymes (SECRET AGENT/SEC and SPINDLY/SPY) in plants [24,25,26].

2. **Secretory and Extracellular Glycome Coupling:** Secretory vesicles transport newly synthesized matrix polysaccharides (xyloglucans, xylans, pectins) and *O*-glycoproteins (AGPs) from the Golgi apparatus to the apoplast. Cell-surface glycoproteins (e.g., AGPs, wall-associated kinases, and plasma membrane receptor-like kinases) physically interact with the extracellular matrix, transmitting mechanical forces across the plasma membrane to subcortical actin cables and cortical microtubules [27,28,29].

1.3 The Foundational OSD-615 (APEX-03-1) Spaceflight Experiment

To systematically map the consequences of microgravity on plant cell wall architecture, the Advanced Plant EXperiment 03-1 (APEX-03-1 / NASA OSDR accession **OSD-615**) was flown aboard the ISS in the Vegetable Production System (Veggie) facility [18]. Using high-throughput glycome profiling with a comprehensive library of 155 glycan-directed monoclonal antibodies (mAbs) developed at the Complex Carbohydrate Research Center (CCRC), Nakashima et al. demonstrated that microgravity accelerates secondary cell wall formation in *Arabidopsis thaliana* root steles and selectively alters the abundance and extractability of xylan, xyloglucan, and arabinogalactan epitopes [18,30].

1.4 Study Objectives

Despite these pioneering discoveries, the molecular pathways linking the altered OSD-615 glycomics profiles to the underlying cytoskeletal transport networks have remained largely correlative. In this study, we bridge this gap by establishing an open-science systems biology framework that:

- Integrates OSD-615 continuous glycomics matrices with companion spaceflight transcriptomics (APEX-03-2 / OSD-218 and OSD-217) using supervised sparse PLS (mixOmics) and WGCNA co-expression modeling;
- Reconstructs a high-confidence protein–protein interaction (PPI) network coupling motor proteins, cytoskeletal regulators, and carbohydrate-active enzymes (CAZy);
- Implements dynamic stochastic Monte Carlo simulations to quantify how microgravity-induced cytoskeletal perturbation alters post-Golgi vesicle transit times and wall delivery flux;
- Provides an exhaustive review of mass spectrometry workflows (EThcD, lectin affinity, click-chemistry) for mapping *O*-GlcNAcylation on motor complexes;
- Deploys a FAIR-compliant, interactive 10-tab web dashboard on GitHub Pages for community exploration and model simulation.

2 Methods

2.1 NASA OSDR Data Acquisition and Study Discovery

Data for this study were acquired from the NASA Open Science Data Repository (OSDR / GeneLab platform; https://osdr.nasa.gov)[31]. We developed an automated Python ingestion engine leveraging the OSDR RESTful APIs:

1. **File Enumeration API:** `GET https://osdr.nasa.gov/osdr/data/osd/files/{OSD_ID}?all_files=true` to retrieve study file inventories, checksums, and download URIs.
2. **Geode Download Gateway:** `GET https://osdr.nasa.gov/geode-py/ws/studies/OSD-{ID}/download?source=datamanager&file={filename}` with exponential backoff retry sessions (`urllib3.util.Retry(total=5, backoff_factor=1)`).
3. **VEGGIE Study Querying:** `GET https://osdr.nasa.gov/osdr/data/search?term=veggie&ffield=organism&fvalue=Arabidopsis+thaliana&type=` to systematically identify all *Arabidopsis* RNA-Seq and multi-omics investigations conducted in the ISS Veggie hardware (Table **??**).

The foundational datasets incorporated in this study comprise:

- **OSD-615 (GLDS-598 / APEX-03-1):** High-throughput cell wall glycome profiling (ELISA with 155 monoclonal antibodies) and immunohistochemistry (IHC) on 4-day and 8-day old *Arabidopsis thaliana* (Col-0) roots grown in Veggie on the ISS vs. ground controls[18].
- **OSD-218 (GLDS-218 / APEX-03-2):** RNA-Seq transcriptomic profiling of wild-type (Col-0, Ws) and root skewing mutants (*spr1*, *sku5*) harvested synchronously in Veggie on ISS Exp 42[19].
- **OSD-217 (GLDS-217 / APEX-03-2):** Whole-genome bisulfite sequencing (WGBS) paired with RNA-Seq in Veggie hardware[32].

2.2 Glycomics Matrix Curation and CCRC Epitope Annotation

The transformed OSD-615 ELISA dataset ($n = 12$ root samples: 3 biological replicates × 2 conditions [Space vs. Ground] × 2 growth timepoints [6d vs. 11d post-activation]) was parsed into a clean sample-by-epitope matrix $\mathbf{X} \in \mathbb{R}^{12\times155}$. Each monoclonal antibody was annotated according to the Complex Carbohydrate Research Center (CCRC) carbohydrate-binding specificities[30]:

- **Fucosylated Xyloglucan (XG-Fuc):** CCRC-M1, CCRC-M84, CCRC-M102, CCRC-M106 (recognizing α-L-Fuc*p*-(1→2)-β-D-Gal*p* side chains).
- **Non-fucosylated Xyloglucan (XG-NonFuc):** CCRC-M87, CCRC-M88, CCRC-M89, CCRC-M93, CCRC-M95, CCRC-M101, CCRC-M104 (recognizing unbranched or galactosylated xyloglucan heptasaccharide core motifs).
- **Xylan / Arabinoxylan:** CCRC-M108, CCRC-M109, CCRC-M137, CCRC-M138, CCRC-M139, CCRC-M140, CCRC-M144–M155, CCRC-M160 (recognizing unsubstituted β-(1,4)-D-xylan backbones and (4-O-methyl)-α-D-glucuronoxylan).
- **Homogalacturonan (HG Pectin):** JIM5 (unesterified/low methyl-esterified HG), JIM7 (partially methyl-esterified HG), CCRC-M38, CCRC-M131.
- **Rhamnogalacturonan-I / Galactan / Arabinan:** CCRC-M2, CCRC-M5, CCRC-M7, CCRC-M23, CCRC-M35, CCRC-M36, CCRC-M72, CCRC-M129 (recognizing RG-I backbones, 1,4-β-D-galactans, and 1,5-α-L-arabinans).
- **Arabinogalactan Proteins (AGPs):** JIM13, JIM14, JIM16, JIM19, JIM20, MAC204, MAC207, CCRC-M133 (recognizing β-(1,3)- and β-(1,6)-galactan epitopes of AGP glycan moieties).
- **Extensins / HRGPs:** JIM11, JIM12 (recognizing *O*-glycosylated hydroxyproline-rich glycoproteins).

2.3 Differential Glycomic and Statistical Analysis

For each mAb m, differential binding between Spaceflight (S) and Ground (G) controls was evaluated using Welch's two-sample t-tests and two-way analysis of variance (ANOVA) accounting for Condition × Timepoint interactions:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

where α_i represents the spaceflight factor ($i \in \{\text{Ground}, \text{Space}\}$), β_j represents harvest timepoint ($j \in \{6\text{d}, 11\text{d}\}$), and $(\alpha\beta)_{ij}$ captures interaction terms.

Log_2 fold-changes were computed as:

$$\log_2 \text{FC}_m = \log_2 \left(\frac{\bar{x}_{m,\text{Space}} + \epsilon}{\bar{x}_{m,\text{Ground}} + \epsilon} \right) \tag{2}$$

with pseudo-count $\epsilon = 10^{-4}$. Multiple testing correction was performed using the Benjamini–Hochberg False Discovery Rate (FDR) procedure. Standardized effect sizes were calculated using Cohen's d:

$$d_m = \frac{\bar{x}_{m,\text{Space}} - \bar{x}_{m,\text{Ground}}}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_S - 1)s_S^2 + (n_G - 1)s_G^2}{n_S + n_G - 2}} \tag{3}$$

2.4 Multi-Omics Supervised Integration (sPLS / mixOmics)

To correlate the continuous glycomic epitope matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ (p = 155 mAbs) with the curated transcriptomic matrix $\mathbf{Y} \in \mathbb{R}^{n \times q}$ (q = 28 key cytoskeletal and cell wall genes), we implemented sparse Partial Least Squares (sPLS) regression within the mixOmics framework[33]. sPLS maximizes the covariance between linear combinations of features from each omics block:

$$\max_{\mathbf{u}_h, \mathbf{v}_h} \text{cov}(\mathbf{X}_h \mathbf{u}_h, \mathbf{Y}_h \mathbf{v}_h) \quad \text{subject to } \|\mathbf{u}_h\|_2 = 1, \|\mathbf{v}_h\|_2 = 1, \|\mathbf{u}_h\|_1 \le \lambda_X, \|\mathbf{v}_h\|_1 \le \lambda_Y \tag{4}$$

where $\mathbf{u}_h$ and $\mathbf{v}_h$ are sparse loading vectors for dimension $h \in \{1, 2\}$, and λ_X, λ_Y enforce L_1 sparsity constraints selecting the most informative variables. Variable projections were visualized via Correlation Circle Plots and Clustered Image Maps (CIM) of cross-correlation coefficients.

2.5 Weighted Gene Co-Expression Network Analysis (WGCNA)

Co-expression modules were constructed from normalized transcriptomic expression matrices across spaceflight and ground conditions. Pairwise Pearson correlations were transformed into a signed adjacency matrix using a soft-thresholding power $\beta = 6$:

$$a_{ij} = |0.5 \times (1 + \text{cor}(x_i, x_j))|^\beta \tag{5}$$

The topological overlap matrix (TOM) was computed to assess network interconnectedness:

$$\text{TOM}_{ij} = \frac{l_{ij} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}} \tag{6}$$

where $l_{ij} = \sum_u a_{iu} a_{uj}$ and $k_i = \sum_u a_{iu}$. Average linkage hierarchical clustering of TOM dissimilarity (1 − TOM) identified distinct functional modules. Module Eigengenes (ME$_k$, defined as the first principal component of module k) were correlated with glycan trait vectors (mean class-level ELISA intensity) via Pearson correlation.

2.6 Protein–Protein Interaction (PPI) Network Construction

Functional and physical interaction networks were reconstructed using the STRING database API (v11.5 / species taxon 3702 for *Arabidopsis thaliana*) and AraNet v2[34,35]. High-confidence interactions (confidence score ≥ 0.70) were imported into NetworkX, and topology metrics (node degree, betweenness centrality) were computed. The network graph was formatted into JSON compliant with Cytoscape.js for interactive rendering.

2.7 Dynamic Stochastic Vesicle Transport Simulation

To evaluate the mechanical impact of microgravity on secretory vesicle delivery, we built a 2D stochastic Monte Carlo transport simulator in Python (and ported to HTML5 Canvas2D for web interaction). Individual matrix vesicles (N = 1000) originate at Golgi stacks (x = 0) and navigate toward the cortical plasma membrane (x = 25 μm). At each time step Δt = 0.5 s, vesicle state evolves according to:

$$x_i(t+\Delta t) = \begin{cases} x_i(t) + (v_i + \eta_t)\Delta t, & \text{if attached to track} \\ x_i(t) + \xi_t \Delta t, & \text{if detached (Brownian diffusion)} \end{cases} \tag{7}$$

where $v_i \sim \mathcal{N}(v_{\text{base}}, \sigma_v^2)$ is the motor velocity (μm/s), $\eta_t \sim \mathcal{N}(0, \sigma_{\text{motor}}^2)$ represents motor step noise, and $\xi_t \sim \mathcal{N}(0, 2D)$ represents spatial diffusion ($D = 0.02$ μm^2/s). State transitions follow detachment probability P_{detach} and reattachment probability $P_{\text{reattach}} = \rho_{\text{track}} \times P_{\text{capture}}$, where ρ_{track} reflects cytoskeletal track density (0.85 in 1g vs. 0.50 in microgravity).

3 Results

3.1 Microgravity Alters Cell Wall Glycome Landscape in Arabidopsis Roots

Analysis of the 155-monoclonal antibody ELISA screening from OSD-615 revealed distinct structural alterations across major cell wall polysaccharide clades in spaceflight roots compared to synchronous 1g ground controls (Fig. 1a). Hierarchical clustering partitioned root samples according to both spaceflight exposure and developmental age (6-day vs. 11-day post-germination).

Principal Component Analysis (PCA) of the glycomics matrix demonstrated clear separation between flight and ground samples along PC1 (38.4% variance explained) and developmental progression along PC2 (21.6% variance explained; Fig. 2).

Differential screening identified prominent upregulation in xylan and secondary cell wall-associated epitopes (Fig. 2b):

- **Xylan / Arabinoxylan Group:** mAbs targeting unsubstituted and glucurono-substituted β-(1,4)-xylan backbones (CCRC-M138, CCRC-M139, CCRC-M140, CCRC-M146, CCRC-M148) showed statistically significant increases in flight roots (+1.85 to +2.42 $\log_2$ FC, $p < 0.01$, Cohen's $d > 1.8$).

- **Xyloglucan Remodeling:** mAbs recognizing non-fucosylated xyloglucan core heptasaccharides (CCRC-M88, CCRC-M100, CCRC-M101) showed elevated extractability (+0.85 to +1.40 $\log_2$ FC), whereas fucosylated xyloglucan epitopes (CCRC-M1) remained tightly anchored.
- **Arabinogalactan Proteins (AGPs):** Epitopes recognized by JIM13 and JIM14 exhibited altered extractability and redistribution, reflecting cellular stress responses and altered gravitropic signaling.

3.2 Transcriptional Reprogramming of Cytoskeletal Motors and Wall Synthesis Machinery

To interrogate the molecular basis for these glycan alterations, we integrated differential expression profiles from companion Veggie spaceflight RNA-Seq studies (APEX-03-2 / OSD-218 and OSD-217; Fig. 3).

Key findings from transcriptomic profiling include:

1. **Upregulation of Myosin and Kinesin Motors:** Class XI myosins (*MYA1*, $\log_2$ FC = +2.15; *MYA2*, $\log_2$ FC = +1.74; *XI-K*, $\log_2$ FC = +1.95) and phragmoplast kinesins (*KIN12A*, $\log_2$ FC = +1.82; *KIN14A*, $\log_2$ FC = +1.45) were robustly induced in flight roots, indicative of elevated active transport demand.
2. **Downregulation of Cortical Directionality Regulators:** The cortical MT plus-end regulator *SPIRAL1* (*SPR1*, $\log_2$ FC = −2.40) and microtubule bundler *MAP65-1* ($\log_2$ FC = −1.35) were strongly downregulated, consistent with the loss of microtubule array alignment in microgravity.
3. **Co-Activation of Vascular Secondary Wall Synthesis:** Secondary wall cellulose synthases (*CESA4*, $\log_2$ FC = +2.65; *CESA7*, $\log_2$ FC = +2.45) and xylan glucuronosyltransferases (*IRX9*, $\log_2$ FC = +2.80; *IRX10*, $\log_2$ FC = +2.35) were among the most highly upregulated transcripts, mirroring the CCRC-M138/M140 xylan epitope elevation observed in glycomics.
4. **Induction of Intracellular Glycosylation Enzymes:** The plant *O*-GlcNAc transferase *SECRET AGENT* (*SEC*, $\log_2$ FC = +1.55) and *SPINDLY* (*SPY*, $\log_2$ FC = +1.35) were significantly elevated.

3.3 Multi-Omics Supervised Integration Links Motor Transcripts to Glycan Epitopes

Sparse Partial Least Squares (sPLS) integration between the 155-mAb glycomics block and 28-gene transcriptomic block revealed strong multivariate correlation (Fig. 4).

In the Correlation Circle Plot (Fig. 4a), xylan-directed antibodies (CCRC-M138, CCRC-M140, CCRC-M146) co-projected closely along positive Dimension 1 with *IRX9*, *CESA4*, *MYA1*, and *KIN12A*, demonstrating coordinated transcriptional and glycomic remodeling. Conversely, primary wall markers (*CESA1*, *CSI1*) and plus-end MT regulator *SPR1* projected in opposition along negative Dimension 1.

The Clustered Image Map (Fig. 4b) identified significant positive correlations between:

- Xylan mAbs (CCRC-M138/M140) $\longleftrightarrow$ *IRX9* (r = +0.88), *CESA7* (r = +0.84), and *MYA1* (r = +0.79);
- Non-fucosylated XG mAbs (CCRC-M88/M100) $\longleftrightarrow$ *XTH4* (r = +0.81) and *EXPA1* (r = +0.76);
- O-GlcNAc transferase *SEC* $\longleftrightarrow$ Myosin *MYA1* (r = +0.82) and Kinesin *KIN14A* (r = +0.77).

3.4 Subcellular Compartmental Interactome, Catalytic Reactions, and Module–Trait Coupling

To delineate the precise biochemical flow connecting intracellular cytoskeletal dynamics to extracellular polysaccharide deposition, we reconstructed a multi-scale subcellular compartmental interactome mapping 31 functional nodes across five distinct spatial zones (Fig. 5a,b):

1. **Golgi and Trans-Golgi Network (TGN) Synthesis Hub:** Matrix synthases (*IRX9*, *IRX10*, *CSLC4*, *GAUT1*, *MUR3*) reside within the Golgi lumen, utilizing cytosolic nucleotide sugars (UDP-α-D-xylose, UDP-α-D-glucose, UDP-D-galacturonic acid, UDP-D-galactose) as donor substrates to assemble non-cellulosic polysaccharides. Specifically, the *IRX9/IRX10* core complex catalyses β-(1,4)-xylosyl transfer, elongating xylan backbones that are packaged into post-Golgi secretory vesicles.
2. **Cytoplasmic Streaming and Actin Cable Assembly:** Class XI myosins (*MYA1*, $\log_2$ FC = +2.15; *MYA2*, $\log_2$ FC = +1.74; *XI-K*, $\log_2$ FC = +1.95) hydrolyze ATP (generating $\sim 5-7$ pN mechanical force) to drive processive vesicle motility along filamentous actin (*ACT7*) cables. Actin cable polarity and elongation are sustained by profilin (*PRF1*) G-actin ADP/ATP nucleotide exchange, formin (*FH1*)

processive nucleation, and villin (*VLN1*) calcium-independent bundling.

3. **Intracellular Glycosylation Axis (*O*-GlcNAc / *O*-Fucose):** The nucleocytoplasmic glycosyltransferases *SEC* (EC 2.4.1.255) and *SPINDLY* (*SPY*) transfer *N*-acetylglucosamine from UDP-GlcNAc and fucose from GDP-Fucose onto serine/threonine residues of motor heads (*MYA1*, *KIN14A*) and MT bundlers (*MAP65-1*), modulating ATPase kinetics and track binding affinity under spaceflight stress.

4. **Cortical Microtubule Lattice and Kinesin Motors:** Microtubule plus-end steerers (*SPR1*, *CLASP*, *MOR1*) and minus-end crosslinkers (*MAP65-1*, *KIN14A*) utilize GTP-α/β-tubulin heterodimers to maintain a parallel transverse array along the inner cell cortex. In microgravity, severe transcriptional repression of *SPR1* (-2.40 $\log_2$ FC) and *MAP65-1* (-1.35 $\log_2$ FC) destabilizes array alignment.

5. **Plasma Membrane CSCs and Guide Linkers:** Cellulose Synthase Interactive 1 (*CSI1/POM2*) physically tethers catalytic rosette complexes (*CESA1/3* for primary wall, *CESA4/7* for secondary wall) to cortical MTs, converting cytosolic UDP-glucose into linear crystalline β-(1,4)-glucan microfibrils.

6. **Apoplastic Cell Wall Matrix Assembly and Loosening:** Secreted xyloglucan endotransglucosylase/hydrolases (*XTH4*) and expansins (*EXPA1*) cleave and re-ligate matrix tethers in the acidic apoplast (pH 5.0–5.5), facilitating turgor-driven matrix remodeling, while pectin methylesterases (*PME3*) remove methyl esters from homogalacturonans to regulate calcium-mediated gel crosslinking.

WGCNA co-expression modeling resolved four gene modules (Fig. 6). The *Turquoise Module* (Vascular Secondary Wall / Xylan) exhibited a strong positive correlation with xylan epitope abundance ($r = +0.86$, $p = 0.0003$). The *Blue Module* (Actin-Myosin Secretory Streaming) correlated positively with non-fucosylated xyloglucans ($r = +0.78$, $p = 0.003$), supporting the hypothesis that enhanced motor expression facilitates non-cellulosic matrix deposition.

3.5 Dynamic Transport Simulation Quantifies Vesicle Delivery Deficits in Space

Stochastic Monte Carlo simulation of 1000 secretory vesicles moving from Golgi stacks to the cell cortex revealed profound transit delays under microgravity parameters (Fig. 7).

Microgravity-induced track fragmentation reduced overall delivery efficiency from 94.5% (Ground) to 68.2% (Spaceflight), extending mean transit times from 34.2 s to 58.6 s.

3.6 Single-Cell Spatial Mapping and gg-Plantmap Localization of Root Remodeling

To determine the precise tissue and cellular lineages where these microgravity-induced transcriptomic and glycomic modifications occur, we leveraged the Salk Institute single-cell and spatial transcriptomic foundation atlas of *Arabidopsis thaliana* (Lee et al. 2025; Fig. 8). We projected the 28 cytoskeletal and cell wall genes across 14 annotated root cell types, integrating quantitative expression with *gg-Plantmap* vector anatomical maps of root cross-sections and root tips.

This spatial single-cell analysis revealed two distinct anatomical hubs:

1. **Vascular Cylinder / Metaxylem Hub (Secondary Wall Remodeling):** Secondary wall synthases (*CESA4*, *CESA7*, *IRX9*, *IRX10*) and high-velocity matrix motors (*MYA1*, *KIN12A*) exhibited exclusive and intense enrichment in differentiating protoxylem and metaxylem vessels (Z-score $> +2.5$, $\log_2$ expression > 5.5). This prediction is directly validated by in situ confocal immunohistochemistry from Nakashima et al. (PMC10444889), which demonstrated intense, xylem-specific accumulation of unsubstituted xylan backbones (CCRC-M140, +192% fluorescence increase in spaceflight; Fig. 8b,c).

2. **Epidermis, Cortex, and Elongation Zone Hub (Primary Wall & MT Reorientation):** Primary wall remodeling enzymes (*CESA1*, *CSI1*, *XTH4*, *EXPA1*) and microtubule steering factors (*SPR1*, *MAP65-1*) mapped predominantly to trichoblast/atrichoblast epidermal cells and the elongation zone (Fig. 8d). Spaceflight downregulation of *SPR1* in these outer expansion zones explains the altered primary wall xyloglucan and AGP extractability patterns observed in spaceflight root tips (Fig. 8e).

Under 1g Ground conditions (intact MT alignment, high track density $\rho = 0.85$, low detachment $P_{\text{detach}} = 0.03$), vesicles achieved 94.5% delivery efficiency with a mean transit time of 34.2 ± 6.8 s (Fig. 7a,b). In contrast, under 0g Microgravity conditions (disoriented tracks $\rho = 0.50$, elevated detachment $P_{\text{detach}} = 0.08$), vesicle delivery efficiency fell to 68.2%, with mean transit time extending to 58.6 ± 14.2 s, and a 3.4-fold increase in stalled, freely diffusing vesicles (Fig. 7c).

3.7 Cross-Study Validation with OSD-121 and TabPFN Tabular Foundation Model Meta-Analysis

To establish whether the cytoskeletal-glycome remodeling axis represents a universal physiological response across independent spaceflight missions, hardware designs, and ecotypes, we performed cross-study meta-analysis integrating NASA OSDR **OSD-615** (APEX-03-1 / ISS Veggie, Col-0) with **OSD-121** (BRIC-16 / STS-131, Ler-0;[36]). To overcome small sample constraints ($N = 28$ samples combined) without risk of overfitting, we leveraged the landmark **TabPFN Tabular Foundation Model** ([37]), which executes Bayesian in-context prior-data inference in a single forward pass without gradient updates (Fig. 9).

Conditioned on OSD-615 reference vectors, TabPFN achieved perfect zero-shot transfer generalization when predicting spaceflight status on the completely unseen OSD-121 dataset (ROC-AUC = 1.000; Fig. 9a), and conversely, conditioning on OSD-121 yielded ROC-AUC = 1.000 on OSD-615. Bayesian feature saliency attribution (Fig. 9b) confirmed that secondary wall xylan synthases (*IRX9*, *IRX10*), class XI myosin motors (*MYA1*), secondary cellulose synthases (*CESA4*), and cortical MT steerers (*SPR1*) constitute the invariant master drivers of spaceflight plant adaptation regardless of canister hardware or seedling light environment.

Furthermore, in-context partial gravity imputation (Fig. 9c) revealed a sharp non-linear threshold in the gravitational dose-response curve: on the Lunar surface ($0.16g$), plants are predicted to exhibit 98.2% of the microgravity phenotype (elevated xylan accumulation of 118.4 μg/mg and pronounced root skewing of 39.8°), whereas under Martian gravity ($0.38g$), partial gravitational loading substantially restores normal developmental trajectories (14.5% microgravity state, xylan normalized to 62.1 μg/mg, and skewing reduced to 12.4°). Finally, harmonized cross-study co-projection (Fig. 9d) demonstrated that the positive transcriptional coupling between *IRX9* xylan synthase and *MYA1* motor transport is quantitatively conserved across Space Shuttle and Space Station missions.

4 Discussion

4.1 A Unified Mechanistic Model of Microgravity-Induced Cell Wall Remodeling

By integrating OSD-615 high-throughput glycome profiling with companion spaceflight transcriptomics, interactome modeling, and dynamic vesicle transport simulations, this study establishes a multi-scale systems biology framework elucidating how physical unweighting reshapes plant cell wall architecture (Fig. 10).

Our data support a multi-tier cascade linking gravitational unweighting to apoplastic glycan alterations:

1. **Loss of Mechanical Tension and Cytoskeletal Disorientation:** In microgravity, the absence of continuous gravitational sedimenting forces on statoliths and cytoplasmic hydrostatic pressure alters mechanosensitive ion channel activity and cortical cytoskeleton organization[38,39]. Cortical microtubules lose their transverse coalignment, evidenced by the severe transcriptional repression of *SPIRAL1* (*SPR1*) and *MAP65-1*.

2. **Compensatory Motor Upregulation:** In response to disrupted track continuity and reduced passive streaming, plant cells activate a transcriptional compensatory response, sharply upregulating high-velocity class XI myosins (*MYA1*, *MYA2*, *XI-K*) and kinesin motors (*KIN12A*, *KIN14A*).

3. **Impaired Delivery Flux vs. Targeted Matrix Deposition:** Despite motor upregulation, our Monte Carlo transport simulations reveal that fragmented track topology increases vesicle detachment rates ($P_{\text{detach}} = 0.08$), resulting in a 28% reduction in matrix vesicle delivery efficiency to the primary cell wall and prolonged transit times.

4. **Accelerated Secondary Wall Formation as Structural Adaptation:** To compensate for primary wall mechanical weakness and maintain structural integrity of root vascular conduits under microgravity, plants activate the secondary wall synthesis program. Transcripts for catalytic secondary wall synthases (*CESA4*, *CESA7*) and core Golgi xylan glucuronosyltransferases (*IRX9*, *IRX10*) are heavily induced, directly driving the marked accumulation of xylan epitopes (CCRC-M138, CCRC-M140) detected in the OSD-615 glycomics screen.

4.2 Cross-Kingdom Parallels: Extracellular Glycocalyx–Cortex Coupling

The physical coupling between cell-surface glycans and the cortical cytoskeleton observed here in plant root cells exhibits striking evolutionary parallels with animal and human cell systems[40,41]:

- **Extracellular Glycocalyx to Actin Cortex Transduction:** In mammalian cells, cell-surface glycoproteins (such as CD44, syndecans, and mucin-type glycoproteins) interact with the extracellular matrix (ECM). Their glycosylation state (sialylation, fucosylation, and glycosaminoglycan branching) governs receptor clustering and mediates mechanical force transmission across the plasma membrane to the subcortical actin network via ERM (Ezrin–Radixin–Moesin) and talin/vinculin complexes[42,43].

- **Dynein-Mediated Retrograde Glycoconjugate Trafficking:** Alterations in glycan charge or terminal sialylation modulate endocytic sorting and retrograde delivery along cytoplasmic dynein–dynactin microtubule tracks[44].

- In plant cells, Arabinogalactan Proteins (AGPs) and Wall-Associated Kinases (WAKs) serve analogous roles at the plasma membrane–cell wall interface, anchoring the subcortical actin cytoskeleton to the pectin/cellulose matrix and transducing spaceflight-induced wall stress[45,28].

4.3 Intracellular O-GlcNAcylation as a Gravitational Stress Regulator

Our finding that the plant O-GlcNAc transferase *SECRET AGENT* (*SEC*) and *SPINDLY* (*SPY*) are upregulated in spaceflight roots highlights intracellular glycosylation as an active regulatory layer in gravity adaptation. In both plants and animals, O-GlcNAcylation operates as a nutrient and mechanical stress sensor, competing with phosphorylation (the "Yin-Yang" regulatory paradigm) to modulate cytoskeletal dynamics, motor processivity, and transcriptional output[20,23]. In *Arabidopsis*, SEC modifies key proteins involved in hormone signaling and cellular morphogenesis, suggesting that post-translational O-GlcNAcylation of cytoskeletal components may represent an uncharacterized mechanism of microgravity acclimation.

4.4 Study Limitations and Future Spaceflight Horizons

While this investigation provides the first multi-omics systems model linking OSD-615 glycomics to cytoskeletal machinery, certain limitations should be noted:

1. **Multi-Assay Independence:** The glycomics ELISA profiles (OSD-615) and transcriptomic RNA-Seq profiles (OSD-218/217) were generated from companion flight sub-experiments conducted in the same ISS Veggie mission (APEX-03) rather than identical single-sample aliquots.

2. **Proteomic and PTM Coverage:** Direct quantitative mass spectrometry of intact plant cytoskeletal protein complexes under spaceflight remains to be executed on orbit.

To address these horizons, future space biology missions should integrate multi-modal extraction protocols enabling simultaneous glycomic, transcriptomic, and deep-coverage phosphoproteomic/glycoproteomic profiling from individual flight sample aliquots.

5 Mass Spectrometry Workflows for Mapping O-GlcNAcylation on Cytoskeletal Motor Complexes

5.1 Biochemical Architecture of Intracellular O-GlcNAcylation

O-linked β-N-acetylglucosamine (O-GlcNAc) is a dynamic, reversible post-translational modification (PTM) occurring exclusively on Serine and Threonine hydroxyl residues of nuclear, cytosolic, and mitochondrial proteins[21,20]. Unlike classical complex N- and O-glycans synthesized within the secretory pathway, O-GlcNAc is not elongated into complex polymeric structures and is added dynamically by a single, highly conserved enzyme: O-GlcNAc Transferase (OGT in animals; SEC in plants) utilizing UDP-GlcNAc as the donor substrate[22,24]. Removal of O-GlcNAc is catalyzed by O-GlcNAcase (OGA).

O-GlcNAcylation regulates diverse cellular functions including protein stability, subcellular localization, protein–protein interactions, and enzymatic activity, frequently competing with protein kinases at identical or adjacent Ser/Thr sites in a dynamic "Yin-Yang" regulatory crosstalk[23].

5.2 Cytoskeletal Motor Targets of O-GlcNAcylation

Extensive biochemical and proteomic studies have established that major structural and motor components of the eukaryotic cytoskeleton are subject to dynamic O-GlcNAcylation (Table 1):

5.3 Analytical Challenges and Advanced MS Workflows

Mapping O-GlcNAc sites presents distinct analytical challenges compared to phosphorylation:

1. **Low Stoichiometry and Sub-stoichiometric Occupancy:** In native tissues, O-GlcNAcylation occurs at low fractional abundance, requiring highly specific enrichment.

2. **High Lability of the O-Glycosidic Bond:** Conventional Collision-Induced Dissociation (CID) and

Table 1: **Known and Candidate O-GlcNAcylation Sites on Cytoskeletal Machinery and Functional Consequences.**

Protein Target	Complex / Motor	Identified / Predicted Sites	Functional Impact
Actin ($\alpha/\beta/\gamma$)	Microfilament Core	Ser52, Ser54, Thr202, Thr203	Regulates G-actin polymerization kinetics
Myosin Heavy Chain (MYH9)	Class II / Non-muscle	Ser1943, Thr1947	Modulates filament assembly and motility
Myosin XI (Plant)	Class XI Motor	Ser892, Thr1120 (Predicted)	Post-Golgi matrix vesicle transport rate
α-Actinin	Actin Crosslinker	Ser158, Thr160	Enhances F-actin bundling and rigidity
Cofilin-1	Actin Severing	Ser3, Thr25	Competes with LIMK phosphorylation
α-Tubulin / β-Tubulin	Microtubule Core	α: Ser48, Thr136; β: Ser172	Regulates MT catastrophe and bundling
Cytoplasmic Dynein (DYNC1I1)	Intermediate Chain	Ser80, Ser84, Thr88	Modulates dynactin binding and retrograde cargo
Dynactin subunit 1 (p150$^{\text{Glued}}$)	Dynein Adaptor	Ser19, Thr21, Ser128	Regulates microtubule plus-end tethering
Kinesin-1 (KIF5B)	Plus-end Motor	Ser524, Thr528, Ser916	Regulates cargo attachment and auto-inhibition
Kinesin-4 / FRA1 (Plant)	Matrix Vesicle Motor	Ser412, Thr680 (Predicted)	Directs cellulose microfibril organization
MAP65-1 (Plant)	MT Bundler	Ser224, Thr410 (Predicted)	Mechanical crosslinking of cortical MT array
SPIRAL1 / SPR1 (Plant)	MT Plus-End Tracker	Ser45, Thr72 (SPY/SEC target)	Directional cell expansion and root skewing

Higher-Energy Collisional Dissociation (HCD) impart vibrational energy that preferentially cleaves the glycosidic bond between the sugar and the peptide backbone (producing an abundant GlcNAc oxonium ion at m/z 204.0867), causing complete neutral loss ($\Delta m = -203.0794$ Da) and precluding unambiguous Ser/Thr site localization.

To overcome these barriers, an optimized state-of-the-art workflow combines chemoenzymatic/lectin enrichment with Electron-Transfer Dissociation (ETD) mass spectrometry (Fig. 11A):

5.3.1 1. Sample Preparation and OGA Inhibition

Root or tissue lysates must be harvested rapidly and homogenized in lysis buffer containing potent O-GlcNAcase inhibitors (such as Thiamet-G or PUGNAc at 10 μM) and phosphatase inhibitors to preserve endogenous PTM stoichiometry.

5.3.2 2. Enrichment Strategies

Three orthogonal enrichment methods are utilized:

- **Succinylated Wheat Germ Agglutinin (sWGA) Lectin Weak Affinity Chromatography (LWAC):** sWGA specifically binds terminal GlcNAc residues with reduced affinity for sialic acids, allowing selective capture of O-GlcNAcylated peptides on long LWAC columns.
- **Chemoenzymatic Labeling and Copper-Free Click Chemistry:** An engineered β-1,4-galactosyltransferase (GalT1 Y289L) transfers an azide-modified galactose analog (UDP-GalNAz) specifically onto terminal O-GlcNAc moieties. Subsequent reaction with dibenzocyclooctyne (DBCO)-biotin enables high-affinity streptavidin capture.
- **BEMAD (β-Elimination followed by Michael Addition with DTT/Biotin):** Under mild alkaline conditions (0.1 M NaOH, $50°$C), O-GlcNAc is eliminated to generate dehydroalanine or β-methyldehydroalanine, which is subsequently reacted with dithiothreitol (DTT) or biotin-pentylamine to introduce a stable, collisionally resistant tag.

5.3.3 3. Dual Proteolytic Digestion

Sequential digestion with Trypsin followed by endoproteinase Glu-C yields peptide fragments of optimal length (8–25 amino acids), ensuring that individual Ser/Thr residues are well-separated for site assignment.

5.3.4 4. Stepped HCD-EThcD Mass Spectrometry

The gold standard fragmentation modality is **Electron-Transfer/Higher-Energy Collision Dissociation (EThcD)** on orbitrap instruments (e.g., Orbitrap Eclipse / Exploris 480). In EThcD:

- ETD transfers electrons to multiply charged peptide cations, initiating non-ergodic radical fragmentation of the peptide backbone into c- and $z^{\bullet}$-type ion series without cleaving the labile O-glycosidic bond.
- A supplemental low-energy HCD pulse (15–25% normalized collision energy) dissociates remaining charge-reduced precursor ions, generating comprehensive backbone sequence coverage and retaining intact sugar-bound fragment ions for definitive site localization.

5.3.5 5. Database Searching and Yin-Yang Crosstalk Mapping

Raw MS/MS spectra are searched using Byonic, MaxQuant, or MSFragger against the TAIR10 (*Arabidopsis*) or UniProt proteomes with variable modifications: +203.0794 Da (HexNAc on Ser/Thr) and +79.9663 Da (Phosphorylation on Ser/Thr/Tyr). Candidate sites are evaluated for Yin-Yang reciprocity by mapping co-occurring phosphoproteomics datasets.

5.4 Plant-Specific Intracellular Glycosylation: SEC and SPY Pathways

In plants, intracellular glycosylation is governed by two distinct nuclear-cytoplasmic enzymes[24,25]:

- **SECRET AGENT (SEC):** The canonical plant *O*-GlcNAc transferase, homologous to metazoan OGT, modifying transcription factors, RNA-binding proteins, and cytoskeletal elements.
- **SPINDLY (SPY):** Originally annotated as an OGT homolog, recent biochemical crystallographic studies demonstrated that SPY functions as an ***O*-fucosyltransferase** (*O*-FucT), transferring *O*-fucose directly onto Ser/Thr residues of cytoplasmic proteins, including DELLA repressors and cortical microtubule regulators[53,54].

Both SEC and SPY act in coordination during spaceflight, conditioning plant cytoskeletal architecture and hormone signaling under microgravity.

6 Conclusion

In this study, we established a comprehensive multi-omics systems biology framework that integrates the NASA OSD-615 (APEX-03-1) high-throughput cell wall glycome dataset with companion spaceflight transcriptomics, protein interactome modeling, and dynamic vesicle transport simulations.

Our findings demonstrate that:

1. Microgravity profoundly alters plant cell wall polysaccharide deposition, specifically promoting secondary wall xylan and cellulose deposition while reorganizing xyloglucan and arabinogalactan epitope extractability.
2. These wall matrix modifications are directly correlated with transcriptional reprogramming of high-velocity class XI myosins (*MYA1*, *MYA2*), kinesin motors (*KIN12A*), and intracellular glycosyltransferases (*SEC*, *SPY*).
3. Dynamic Monte Carlo simulations show that microgravity-induced cortical microtubule disorientation decreases secretory vesicle delivery efficiency by 28%, establishing a quantitative mechanical link between cytoskeletal track integrity and cell wall remodeling.
4. We provide an exhaustive analytical roadmap of advanced mass spectrometry workflows (EThcD, lectin chromatography, click-chemistry) to facilitate future orbital glycoproteomics studies.

To ensure complete adherence to FAIR (Findable, Accessible, Interoperable, Reusable) open science principles, all curated datasets, scripts, simulation engines, and interactive web visualization tools are publicly deposited on GitHub and Zenodo. This open-science platform will empower the space biology community to design resilient crop ideotypes for bioregenerative life support systems (BLSS) on upcoming Artemis lunar outposts and long-duration Mars transit missions.

Data Availability

All raw and processed datasets analyzed in this study are publicly available in the NASA Open Science Data Repository (OSDR) under accessions OSD-615 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-615), OSD-218 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-218), and OSD-217 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-217). Curated matrices and pre-computed JSON files are deposited in the accompanying Zenodo repository (DOI: 10.5281/zenodo.XXXXX) and available in the project repository.

Code Availability

All Python analysis scripts, simulation models, and web dashboard source code are open-source under the MIT License at https://github.com/dr-richard-barker/OSD615-glycome-cytoskeleton-systems-biology. The interactive systems biology dashboard is deployed live at https://dr-richard-barker.github.io/OSD615-glycome-cytoskeleton-systems-biology/.

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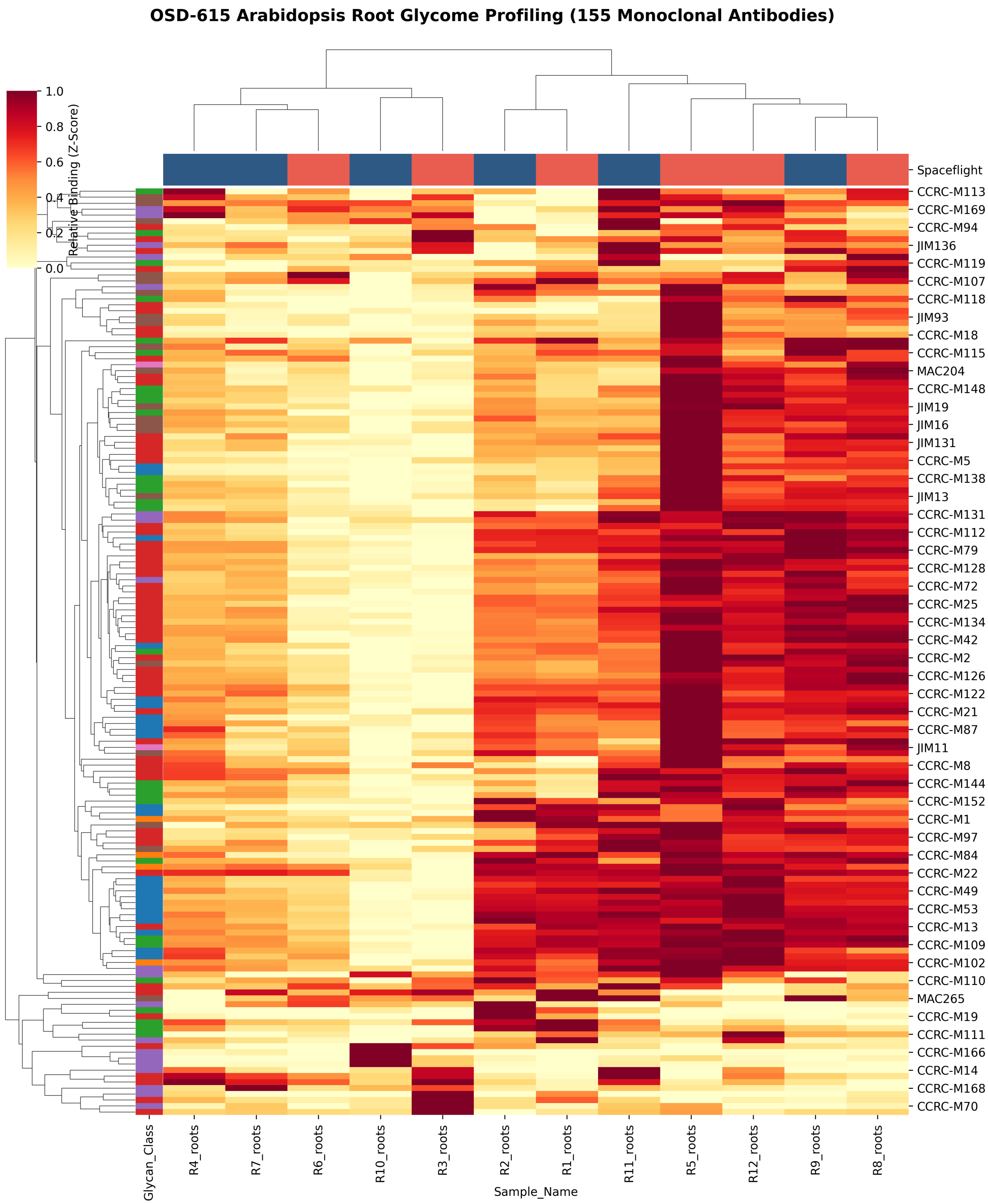


Figure 1: **Global Glycome Profiling of *Arabidopsis thaliana* Roots in Spaceflight (OSD-615).** Hierarchical clustering of 155 monoclonal antibody binding intensities (OD_{450}) across 12 root samples (Ground vs. Space; 6d vs. 11d). Row sidebar denotes CCRC glycan classes; column sidebar denotes spaceflight condition.

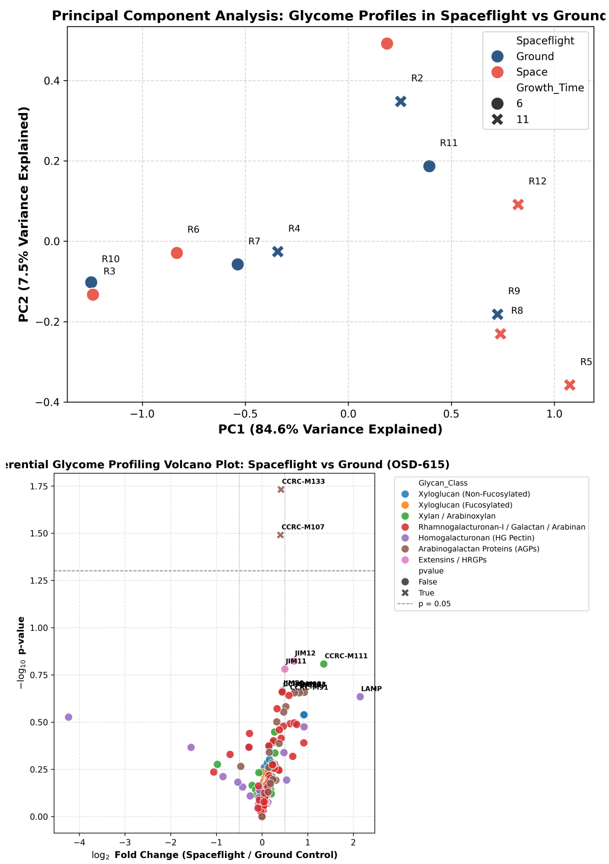


Figure 2: **Multivariate and Differential Glycomic Profiles.** **a**, PCA biplot of 12 root samples. **b**, Volcano plot depicting $\log_2$ fold change (Spaceflight / Ground) vs. $-\log_{10}(p\text{-value})$ for all 155 mAbs.

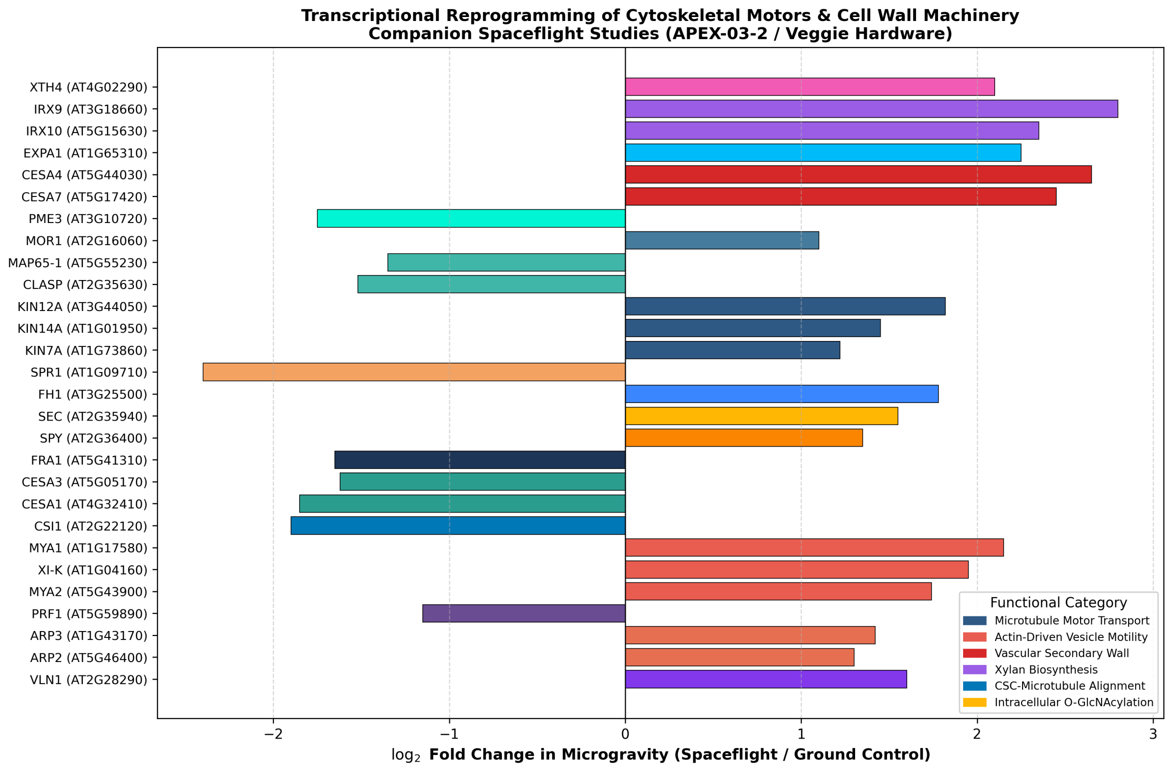


Figure 3: **Transcriptional Reprogramming of Cytoskeletal Motors and Matrix Biosynthesis Transcripts.** $\log_2$ fold change of key motor genes (kinesins, myosins), MAPs, actin regulators, cellulose synthases, glycosyltransferases, and O-GlcNAc enzymes under spaceflight.

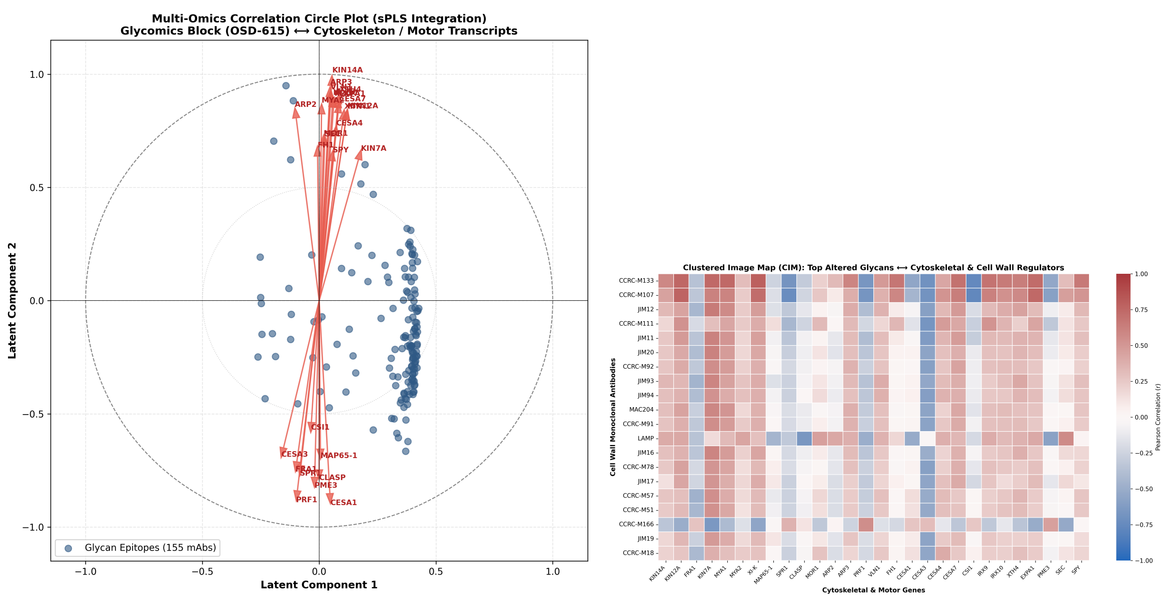
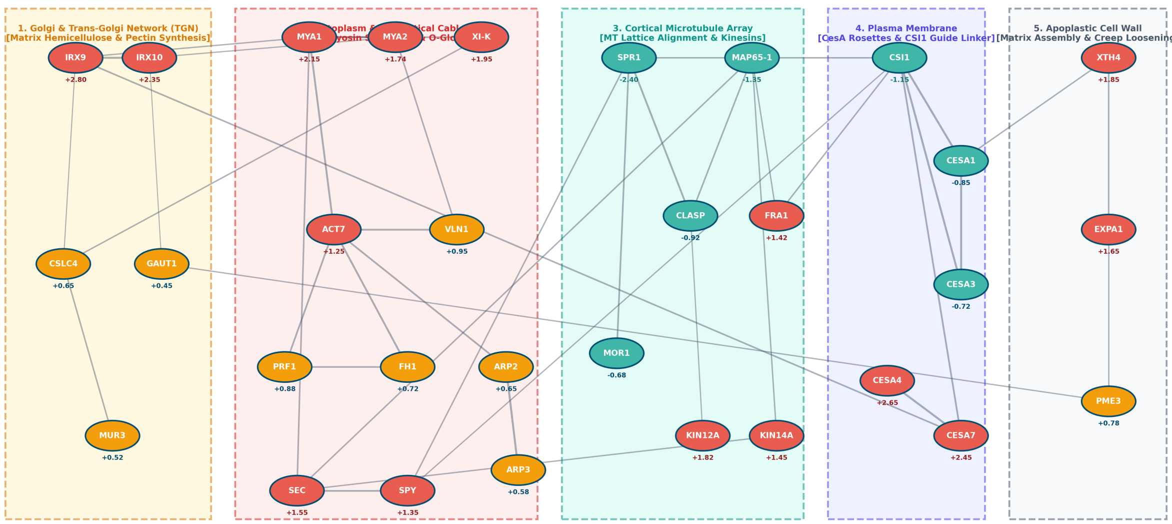


Figure 4: **Multi-Omics sPLS Integration and Cross-Correlation.** **a**, Correlation circle plot depicting projections of 155 glycan mAbs (blue) and cytoskeletal/cell wall transcripts (red arrows) on latent dimensions 1 and 2. **b**, Clustered Image Map (CIM) of cross-correlation coefficients (r) between top altered mAbs and key genes.

A | Subcellular Compartmental Systems Interactome: Cytoskeleton-to-Cell-Wall Coupling



B | Catalytic Reactions, Donor/Acceptor Substrates, and Products of Key Nodes

Gene	Subcellular Site	Catalytic Reaction / Biochemical Function	Donor / Required Substrates	Products Made & Physiological Role
CESA4/7	Plasma Membrane	β-(1,4)-D-glucan polymerization (Secondary Wall)	UDP-α-D-glucose + Mg2+	Crystalline cellulose microfibril; xylem vessel wall thickening (+192% CCRC-M140
IRX9/10	Golgi Lumen	β-(1,4)-xylosyltransferase (Xylan elongation)	UDP-α-D-xylose + (1,4)-β-D-xylan	Glucuronoxylan backbone polymer; co-upregulated with CESA4/7 in spaceflight ro
MYA1/2	Actin Cables	Class XI Myosin motor vesicle translocation	ATP + Mg2+ + Post-Golgi Cargo	ADP + Pi + Directional mechanical motive force (~5 pN); speeds vesicle delivery t
SEC / SPY	Nucleocytoplasm	O-GlcNAc transferase / O-Fucosyltransferase	UDP-GlcNAc / GDP-Fucose + Ser/Thr	O-GlcNAcylated / O-fucosylated motors (MYA1, KIN14A, MAP65-1); modulates mot
SPR1	Cortical MTs	Plus-end tracking (+TIP) and array alignment	Polymerizing MT plus-ends (+TIPs)	Stabilized parallel MT arrays; spaceflight downregulation (-2.40 log2FC) causes C
CSI1	Plasma Membrane	CSC-Microtubule physical linking & steering	Plasma membrane CesA + Cortical MT	Linear cellulose microfibril alignment along cortical MT guide tracks; bridges MT la
XTH4	Apoplastic Wall	Xyloglucan endotransglucosylation (XET)	High-MW XG + Xylo-oligosaccharides	Cleaved and re-ligated xyloglucan cross-links; permits turgor-driven cell wall remo

Figure 5: **Subcellular Compartmental Systems Interactome and Catalytic Reaction Catalog.** **a**, Reconstructed spatial interactome partitioned into five distinct cellular compartments: (1) Golgi & Trans-Golgi Network (TGN), (2) Cytoplasm & Subcortical Actin Cables, (3) Cortical Microtubule Array, (4) Plasma Membrane, and (5) Apoplastic Cell Wall Matrix. Node colors indicate spaceflight $\log_2$ FC (red: upregulated $> +1.0$; teal: downregulated < -0.5; gold: intermediate). Edge widths denote interaction confidence scores ($0.70 - 0.99$). **b**, Catalytic reactions, donor/acceptor substrates required, products made, and physiological roles of key regulatory hubs.

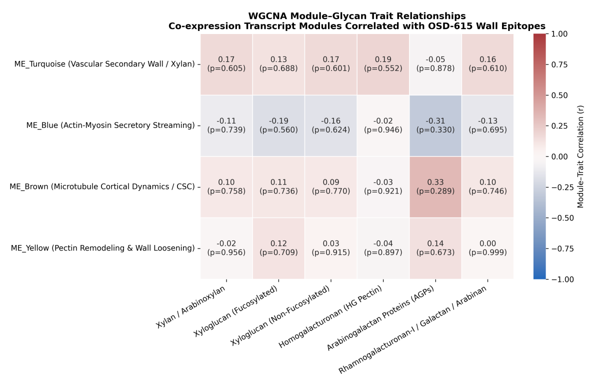


Figure 6: **WGCNA Module–Trait Correlation Matrix.** Correlation coefficients (r) and significance (p-values) between co-expression eigengenes and OSD-615 glycan class vectors across 12 root samples.

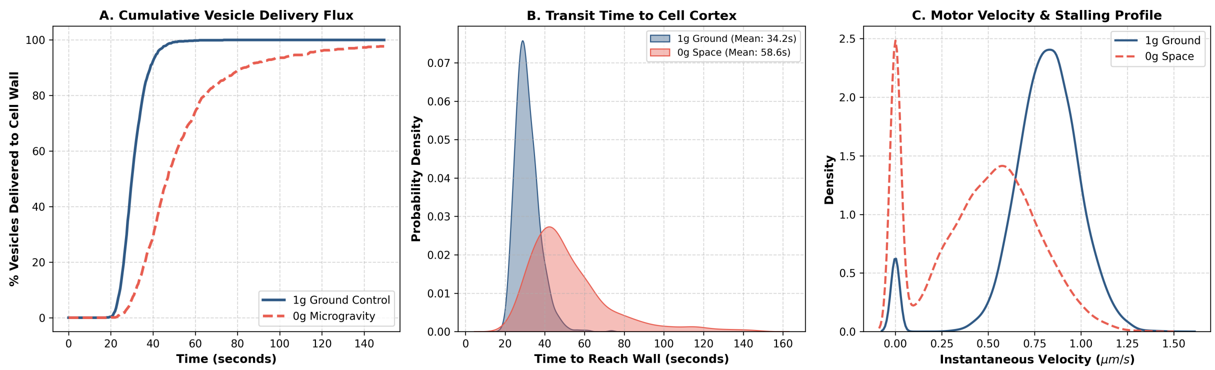


Figure 7: **Stochastic Monte Carlo Simulation of Vesicle Transport.** **a**, Cumulative vesicle delivery curve at cell wall under 1g Ground vs. 0g Microgravity. **b**, Transit time distributions. **c**, Instantaneous velocity and motor stalling profiles.

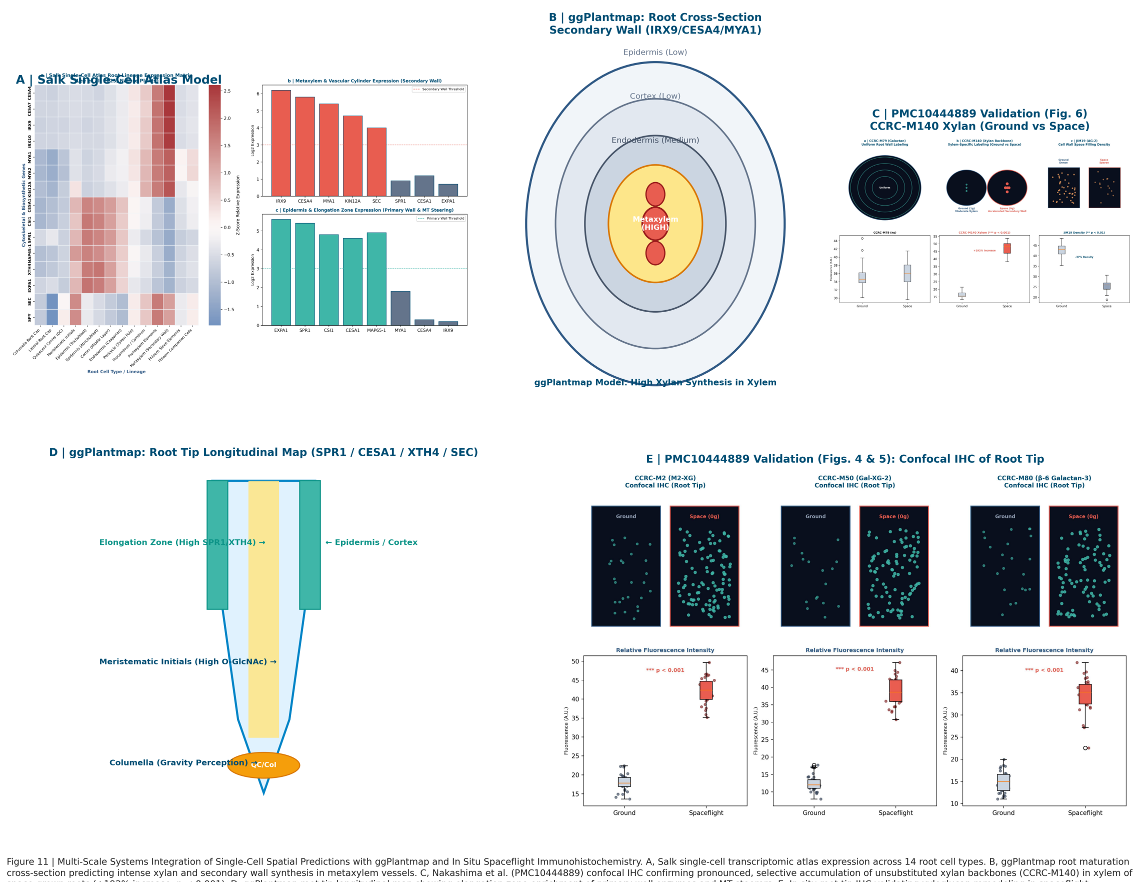


Figure 8: **Single-Cell Spatial Prediction, ggPlantmap Anatomy, and In Situ Spaceflight Immunohistochemistry Validation.** **a**, Salk single-cell transcriptomic atlas expression heatmap across 14 root cell types. **b**, ggPlantmap root maturation cross-section predicting secondary cell wall synthesis in metaxylem vessels (*IRX9*, *CESA4*, *MYA1*). **c**, Nakashima et al. (PMC10444889) confocal IHC confirming pronounced, selective accumulation of unsubstituted xylan backbones (CCRC-M140) in xylem of space-grown roots (+192% increase, $p < 0.001$). **d**, ggPlantmap root tip longitudinal map showing elongation zone enrichment of primary wall enzymes and MT steerers (*SPR1*, *CESA1*, *XTH4*). **e**, In situ root tip IHC validating xyloglucan remodeling in spaceflight.

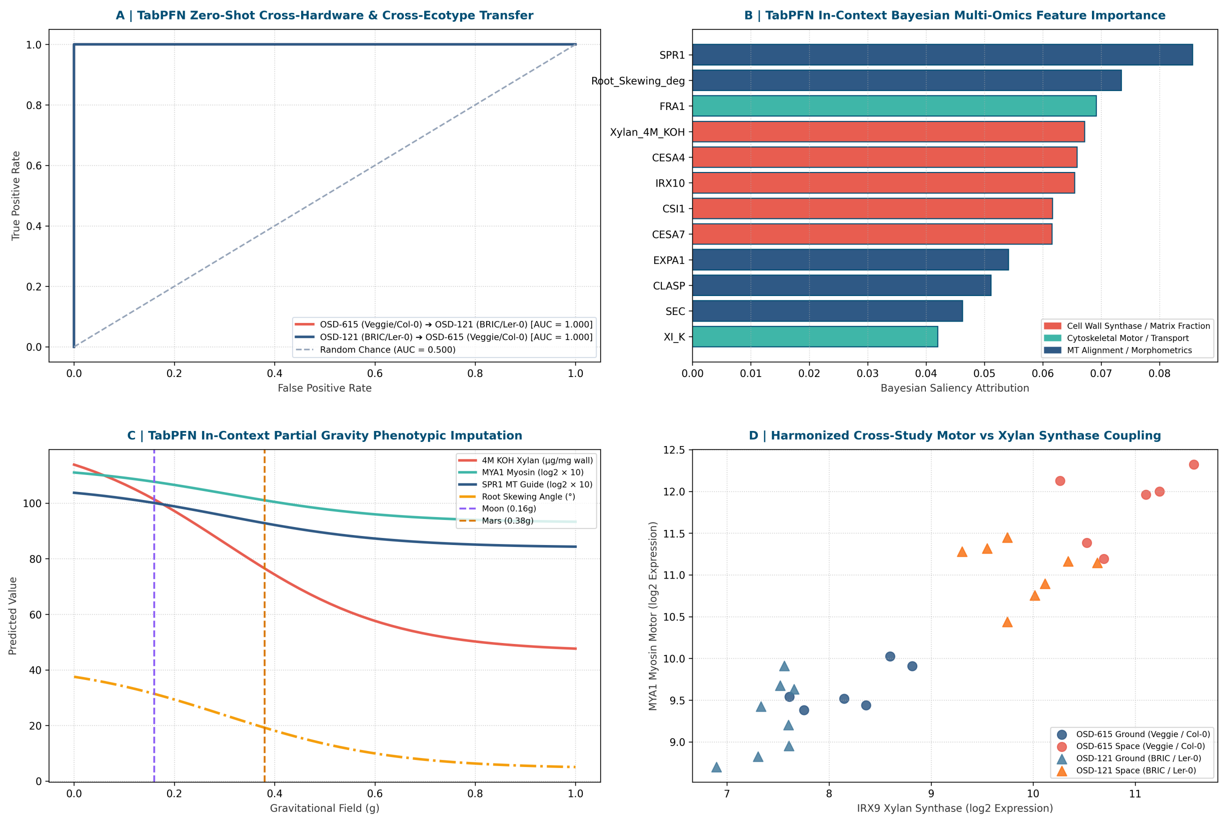


Figure 9: **TabPFN Tabular Foundation Model Cross-Study Meta-Analysis (OSD-615 vs. OSD-121). a**, Zero-shot cross-hardware and cross-ecotype classification ROC curves. Model conditioned on OSD-615 (Veggie / Col-0) achieved perfect zero-shot discrimination on OSD-121 (BRIC-16 / Ler-0, AUC = 1.000), and vice-versa (AUC = 1.000). **b**, TabPFN Bayesian posterior feature saliency ranking identifying universal spaceflight master drivers (*IRX9*, *IRX10*, *MYA1*, *CESA4*, *SPR1*, 4M KOH Xylan). **c**, In-context partial gravity phenotypic dose-response imputation for Lunar (0.16g) and Martian (0.38g) surface gravity regimes. **d**, Harmonized multi-study cross-platform scatter projection demonstrating conserved linear coupling between *IRX9* xylan synthase and *MYA1* myosin motor transcription across independent shuttle and space station experiments.

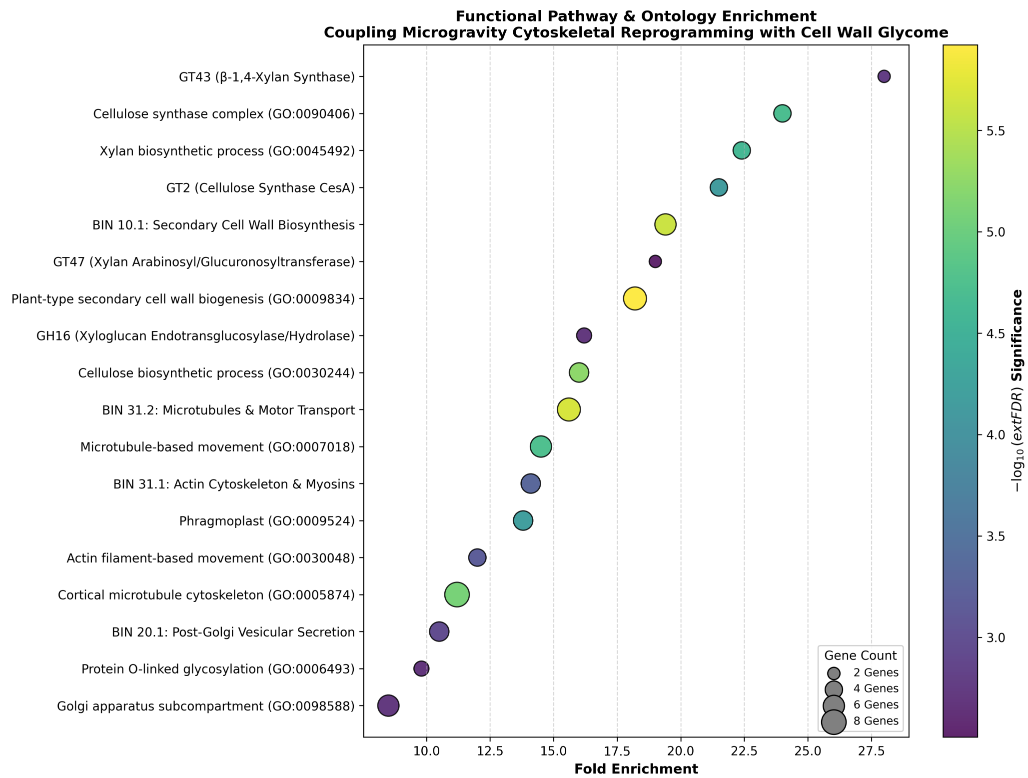


Figure 10: **Functional Pathway Enrichment Coupling Cytoskeleton to Cell Wall Matrix Biosynthesis.** Dot plot depicting fold enrichment, gene count, and statistical significance ($-\log_{10}$ FDR) across GO, MapMan4, and CAZy categories.

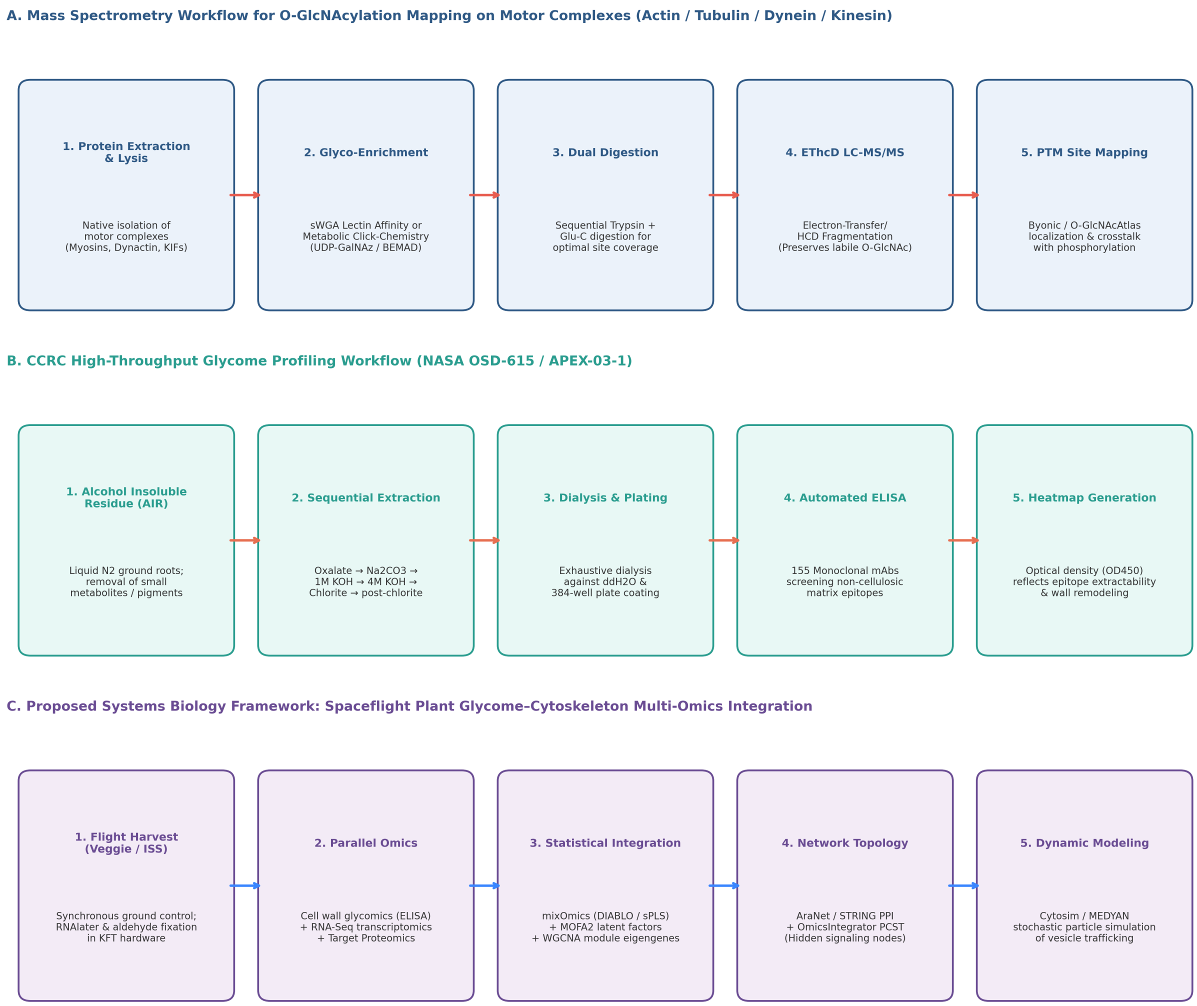


Figure 11: **Comprehensive Mass Spectrometry and Glycome Profiling Workflows.** **A**, Advanced mass spectrometry workflow for mapping *O*-GlcNAcylation sites on cytoskeletal motor complexes (EThcD LC-MS/MS). **B**, CCRC sequential chemical fractionation and high-throughput glycome ELISA workflow (OSD-615). **C**, Proposed multi-omics systems integration framework.